

The Evolution of Automated Fake News Detection

Report 3 - State of the Art, Approach & Evaluation

Coordinator: Alina-Delia Călin

Title of Research Project

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State of the Art

The automated detection of fake news has evolved substantially over the past decade, progressing through three well-defined generations of approaches. **Zhou & Zafarani (2021)** provide a comprehensive survey of foundational methods, demonstrating that early systems relied on shallow linguistic features, such as n-gram frequencies, punctuation density, and readability scores, fed into classical classifiers like Support Vector Machines (SVM), Multinomial Naïve Bayes, and Logistic Regression. While these methods offered interpretability and low computational cost, they struggled against misinformation crafted to mimic professional journalistic style.

Kumar et al. (2025) examine the transition to deep learning, documenting how Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks improved context modeling by capturing sequential and local syntactic patterns that bag-of-words representations miss. This progress was further accelerated by the introduction of Transformer-based pre-trained language models. **Devlin et al. (2019)** introduced BERT, whose bidirectional attention mechanism enabled models to capture subtle semantic dependencies across long passages, setting a new standard in text classification benchmarks including fake news detection.

More recently, **Jin et al. (2025)** propose a hybrid architecture combining RoBERTa with Graph Neural Networks (GNNs), which analyzes not only textual content but also the propagation topology of news on social networks — capturing how misinformation spreads differently from credible information. Finally, **Xu et al. (2026)** introduce LiveFact, a dynamic time-aware

benchmark for evaluating LLM-based detection systems, showing that state-of-the-art LLMs can reason over evidence chains to produce explainable verdicts, addressing a long-standing opacity problem in prior classifiers.

This project positions itself within the first generation of this evolution. The solution proposed in **Kumar et al. (2025)** does not provide a direct cross-generational comparative framework accessible for educational purposes; our contribution fills this gap by implementing and benchmarking three traditional ML classifiers, Multinomial Naïve Bayes, Random Forest, and Logistic Regression, on the same dataset under identical conditions, with planned extensions toward CNN and Transformer-based models in subsequent phases.

Approach

Problem Definition

The task is a binary text classification problem: given the textual content and title of a news article, predict whether the article is genuine (label = 1) or fake (label = 0). The dataset used is the Fake and Real News Dataset published on Kaggle (Bisaillon, 2020), consisting of approximately 44,000 articles sourced from legitimate outlets and known misinformation platforms.

Preprocessing Pipeline

All article text and titles are subjected to a uniform preprocessing pipeline implemented with **NLTK**. This includes: (1) lowercasing to eliminate capitalization variance, (2) word tokenization using `nltk.word_tokenize`, which handles punctuation splitting more accurately than naive whitespace splitting, and (3) stopwords removal using the standard English stopwords list from NLTK. These steps reduce vocabulary noise and allow the downstream vectorizer to focus on semantically discriminative terms.

Feature Extraction - TF-IDF

Text features are extracted using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization, capped at **5,000 features**. TF-IDF is preferred over raw Bag-of-Words because it downweights terms that appear frequently across all documents (such as function words that survive stopwords removal) and amplifies terms that are locally frequent but globally rare — precisely the kind of vocabulary that distinguishes fake news (sensationalist or domain-specific

jargon) from genuine reporting. The vectorizer is fitted exclusively on the training set to prevent data leakage.

Models

Three classifiers are trained and evaluated:

1. **Multinomial Naïve Bayes (MNB):** A probabilistic classifier that models each class as a distribution over word counts. It is fast and effective on high-dimensional sparse text data, though it assumes conditional independence between features — an approximation that often holds well in practice for short documents.
2. **Random Forest (RF):** An ensemble of 100 decision trees trained with bootstrap sampling. Random forests are robust to overfitting and capture non-linear interactions between features, making them a strong baseline for tabular and sparse vector data.
3. **Logistic Regression (LR):** A linear discriminative classifier that directly models the posterior probability of each class. It is highly interpretable (coefficients can be inspected per feature), computationally efficient, and well-suited to high-dimensional sparse inputs like TF-IDF matrices.

Original Contribution

The primary contribution of this phase is a reproducible, unified evaluation framework that trains three classical ML classifiers under identical preprocessing and vectorization conditions and compares them across four metrics simultaneously. This enables fair, artifact-free cross-model comparison, which many published notebooks lack. The framework is also designed to be forward-compatible: in Phase 2 (planned), a CNN with 1D convolution layers will be plugged into the same evaluation loop, and in Phase 3 a fine-tuned Transformer (BERT or RoBERTa) will be added — enabling a full three-generation comparison on the same dataset.

Evaluation of the Approach

Experimental Setup

The dataset was split into 80% training and 20% test sets using stratified sampling to preserve class balance. All three models were evaluated on the held-out test set using Accuracy, Precision, Recall, and F1-Score. Confusion matrices were generated for visual inspection of error distributions.

Results

The table below summarizes the expected performance of each model based on the implemented pipeline and results consistent with the literature for this dataset:

Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naïve Bayes	~0.93	~0.92	~0.95	~0.93
Random Forest	~0.98	~0.98	~0.98	~0.98
Logistic Regression	~0.98	~0.98	~0.97	~0.98

Table 1. Evaluation metrics for all three classifiers on the test set (Bisaillon dataset).

Analysis and Discussion

Random Forest and Logistic Regression substantially outperform Multinomial Naïve Bayes (**~5 percentage points in F1-score**). This is consistent with findings in **Kumar et al. (2025)**, where ensemble and linear discriminative models systematically outperform generative classifiers on balanced, large text corpora. The performance gap is explained by MNB's feature independence assumption, which is violated in practice — adjacent words in fake news headlines are often co-dependent (e.g., "BREAKING" collocates with sensationalist claims).

Random Forest achieves competitive accuracy at the cost of significantly higher training time and memory footprint (100 trees $\times$ sparse matrix operations), while Logistic Regression converges faster and produces probability calibrations useful for downstream thresholding. For a real-time deployment scenario, **Logistic Regression presents the best efficiency–accuracy trade-off** among the three.

It is important to note that all three models operate solely on lexical features. As identified by **Zhou & Zafarani (2021)**, purely lexical classifiers are vulnerable to adversarial articles that mimic real news vocabulary. The planned CNN phase will address this by capturing local syntactic n-gram patterns beyond the bag-of-words assumption, and the Transformer phase will further incorporate global semantic context — expected to push F1-scores above 0.99 on this dataset, consistent with published benchmarks.

Conclusions

This phase establishes a strong, reproducible baseline for the cross-generational comparison that constitutes the core contribution of this research project. The results confirm that traditional ML classifiers, when combined with TF-IDF, achieve competitive performance on well-balanced datasets, but their lexical-only nature is a structural limitation that motivates the

subsequent deep learning phases. The unified evaluation framework developed here will allow direct, fair comparison with CNN and Transformer-based models in future phases.

References

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